

Basics of automation technology

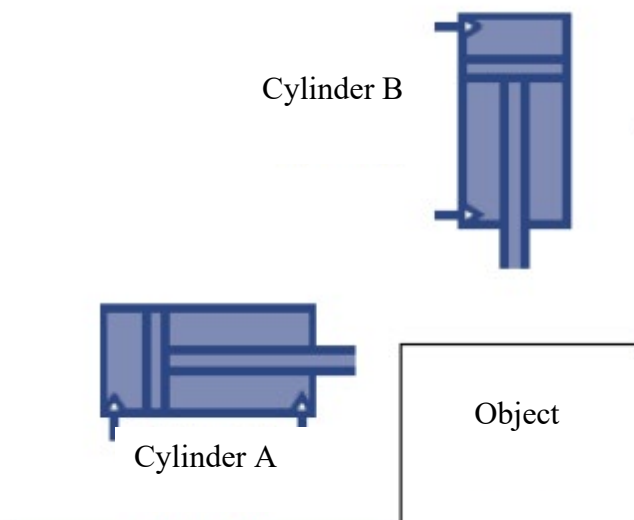
TIA Portal exercise 3

Develop and test a program that implements the operation of the attached application example as shown.

Use the following symbols and addresses:

a0	I0.0	Step1	M1.1
a1	I0.1	Step2	M1.2
b0	I0.2	Step3	M1.3
b1	I0.3	Step4	M1.4
Start	I0.4		
Reset	I0.5		
A+	Q0.0		
A-	Q0.1		
B+	Q0.2		
B-	Q0.3		

Build the program in such a way that each step comes to its own network.



In the initial position, the pistons of machining cylinder B and mounting cylinder A are inside (in the - position). The sequence begins with a start signal, which is obtained from the push-button. At the first mounting cylinder fixes the workpiece. Next, the piston of the

Basics of automation technology

TIA Portal exercise 3

machining cylinder moves out, at which point a hole is drilled in the piece, and further back in. As soon as the piston of the machining cylinder is inside, the fixing cylinder releases the workpiece. After that, the sequence can be restarted from the pushbutton after replacing the piece to be machined.

The task layout can be presented more briefly and clearly by writing the business sequence in the form of a table:

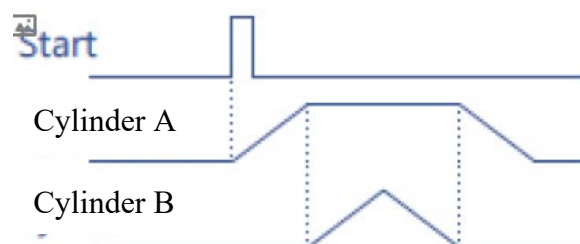
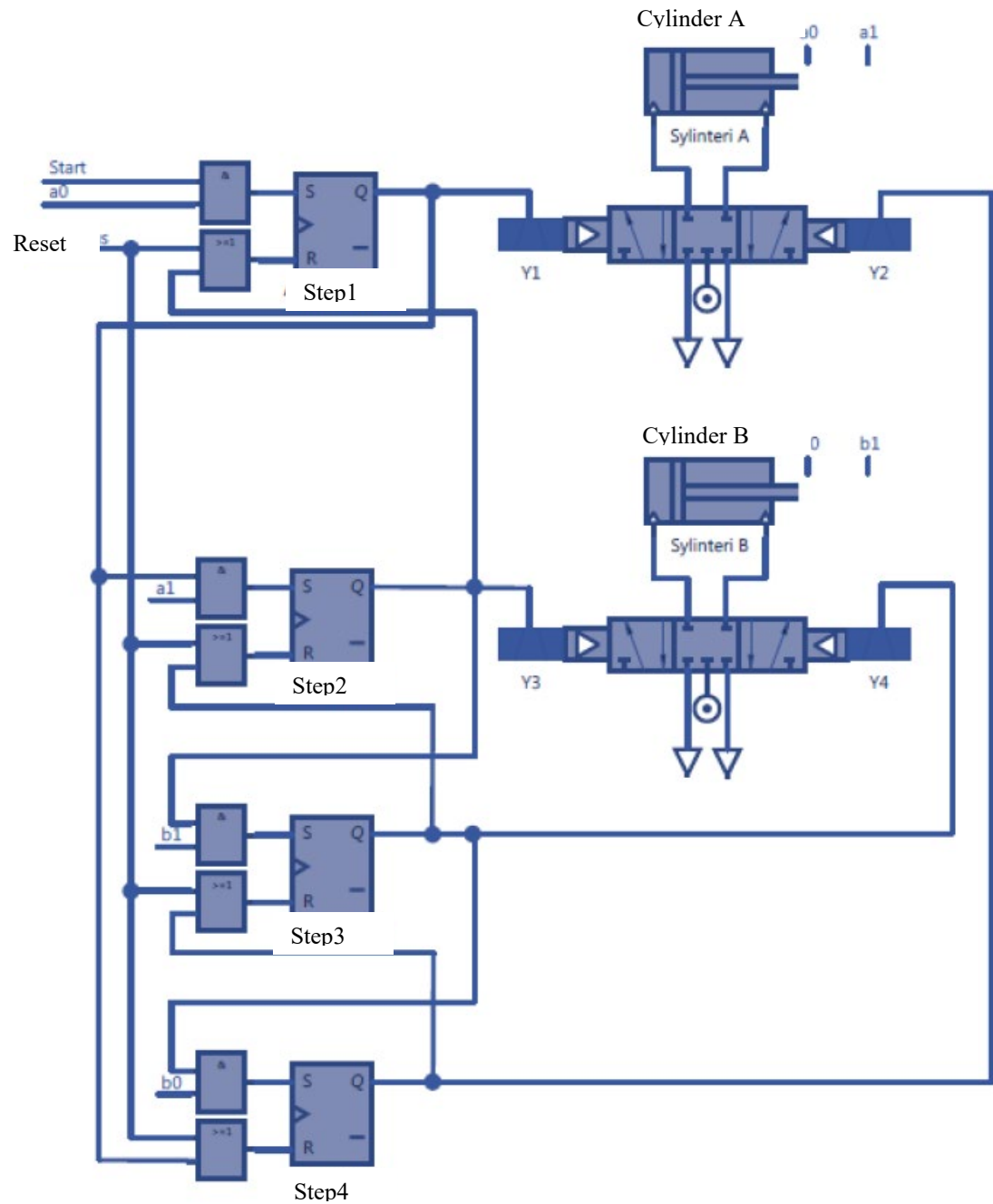
Step		Comment
1	A+	Cylinder A to out position
2	B+	Cylinder B to out position
3	B-	Cylinder B to in position
4	A-	Cylinder A to in position

The cycle consists of four steps. The next step is always started only when a certain condition with the previous step is met.

A graphical representation of a task that contains symbols for individual logical actions (eg. AND, OR, NO, memory), is a detailed description of the control logic. This representation is called a logic diagram. The following illustration shows the signals in the logic diagram combined with the physical devices in the hardware.

Basics of automation technology

TIA Portal exercise 3



Basics of automation technology

TIA Portal exercise 3

When the piston of cylinder A is inside, $a0$ is 1 (true) and when it is outside, $a1$ is 1 (true). The same applies to cylinder B. 5/3 directional valves are used to control both cylinders.

Action Description:

The entire cycle of work can be divided into four steps. For each step, a memory bit is allocated in the program. These memories act as step memories in this context. In the initial state, these memories are reset.

Step 1:

If the start signal has a value of 1 (true) and $a0$ is 1 (the piston in cylinder A is inside), the condition that makes it possible to set the first step memory is met and the value of that memory becomes 1. The cylinder A is activated by $Y1$. The piston of cylinder A moves out (fastening). At the same time, step memory 1 mode 1 (true) is a necessary condition for setting step memory 2.

Step 2:

After cylinder A has moved out, the $a1$ state of the sensor changes to 1.

Kun sylinteri A on liikkunut ulos, anturin $a1$ tila muuttuu 1:ksi. This meets the AND condition, which makes possible the step memory 2 setting. In this case, $Y3$ gets the value 1. The steering body of cylinder B is activated, and the piston of cylinder B moves out (drilling). Step memory 2 value 1 resets step memory 1 using the OR function. It is worth noting that the directional valve of cylinder A is of such a design that the piston in the cylinder, at the end of the ejection, does not make a return movement. After the piston of cylinder B has moved out completely ($b1 = 1$), step 3 is started.

Step 3:

Basics of automation technology

TIA Portal exercise 3

With step 3, cylinder B makes a return movement in with y4 (drilling completed). Step 2 is reset. When the piston of the cylinder has reached its inner position, b0 acquires a value of 1 and triggers step 4.

Step 4:

Step 4 operates through Y2 in such a way that cylinder A also returns in (disconnection). With this movement, the sequence is at an end, and can be started anew.

Test the operation of the program on the simulator by monitoring it. Be careful with the inputs: of course, the boundaries of the cylinders should change correctly in relation to the controls of the cylinders.

Test the operation of the program on the simulator by monitoring it. Be careful with the inputs: of course, the limit sensors of the cylinders should change correctly in relation to the controls of the cylinders. Make a simulation plan in which you mark the steps on how the simulation will be performed.

Return a **PDF report** of the work to the Moodle return box, which presents the program/solution made in the tasks in the form of screenshots and includes a brief description of the challenges and successes of the task.

Also, restore the project as a compressed file. To compress a project, select **Project --> Archive** and select a save folder

